Estimating mitragynine and 7-hydroxymitragynine exposure from self-reported kratom leaf-powder doses

Literature review, pharmacokinetic model, calibration and implementation specification for the *bezKRATOMU* application

Technical report (model version 1)

6 November 2026

Abstract

We describe how the *bezKRATOMU* tapering application will estimate the time course of the two principal opioid-active kratom alkaloids, mitragynine (MG) and its metabolite 7-hydroxymitragynine (7-OH), from the doses (grams and time) that users record. Human pharmacokinetic (PK) studies published 2015–2026 are reviewed. Oral MG is absorbed rapidly ($t_{\max} \approx 1$–$1.5\,\mathrm{h}$) and declines bi-exponentially, with a fast distribution phase over the first $\sim 6\,\mathrm{h}$ and a long terminal phase (reported half-lives 9–68 h depending on dose and sampling duration). 7-OH is formed from MG mainly by hepatic CYP3A4, peaks slightly later ($t_{\max} \approx 1.5$–$2\,\mathrm{h}$) and tracks the parent at a concentration ratio of about 0.1–0.3. We adopt a published linked parent–metabolite two-compartment model (Notario *et al.*, 2026), recalibrate it to the largest dose-ranging study of encapsulated leaf powder (Huestis *et al.*, 2024) with two scale factors ($F_{\mathrm{rel}} = 0.277$, $s_{\mathrm{OH}} = 1.85$), and validate it against independent steady-state data after 15 daily doses. The two analytes are combined into a single potency-weighted index of μ-opioid receptor activity, which is displayed on a relative scale (100 % = the steady-state daily mean at the user's baseline daily amount). Because the model is linear, the curve for any dosing history is an exact superposition of a five-term sum of exponentials, whose coefficients are tabulated here. The result is a population-average estimate, not a measurement: between-subject variability in the source studies is 5–10-fold.

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Scope and caveat. The model estimates what an *average* healthy adult's plasma concentrations would be after the logged doses of a leaf powder containing 1.6 % MG. It does not measure anything, it does not predict effects, craving or withdrawal, and it must not be used to decide when or how much to dose. It is not valid for kratom extracts or 7-OH products.

1 Background pharmacology

Alkaloids. Dried *Mitragyna speciosa* leaf contains dozens of indole and oxindole alkaloids. Mitragynine is the most abundant (about 1–2 % of leaf mass; 1.33 % in the product of Huestis *et al.* [1]; 1.6 % in the product used by our users). 7-Hydroxymitragynine is present only in traces in leaf (<0.01 % in [1]; 0.001 % in our users' product). Other alkaloids (e.g. speciociliatine, paynantheine, speciogynine) reach substantial plasma levels [3] but are less characterised and are not modelled here.

Metabolic origin of 7-OH. In humans, 7-OH plasma exposure after kratom is predominantly the result of hepatic oxidation of MG by CYP3A4. Pretreatment with the selective CYP3A4 inhibitor itraconazole reduced 7-OH $C_{\max}$ by 56 % and $AUC_{0–72}$ by 43 %, while MG $C_{\max}$ rose about 1.5-fold [4]. A physiologically based model reached the same conclusion (CYP3A4 inhibitors reduce 7-OH formation but change MG exposure little) [7]. In mice, 7-OH formed from MG mediates a large part of MG's analgesic effect [9].

Receptor pharmacology. Both compounds are G-protein-biased partial agonists at the human μ-opioid receptor (MOR). In a BRET G-protein activation assay, MG had $EC_{50} = 339 \pm 178$ nM ($E_{\max} = 34$ %) and 7-OH $EC_{50} = 34.5 \pm 4.5$ nM ($E_{\max} = 47$ %) [8, 10], i.e. 7-OH is about tenfold more potent. Binding affinities differ between assays (7-OH K_i 8–47 nM; MG $K_i \approx 230$ nM) [10]. MG also acts at adrenergic and other receptors; these non-opioid actions are ignored in this report.

Protein binding. Both MG and 7-OH are more than 90 % bound in human plasma (equilibrium dialysis) [11]; reported MG binding is 85–97 %. As the unbound fractions are similar, we compare total concentrations.

2 Human pharmacokinetic literature

Table 1: Human PK studies used in this report. Values are means or medians as reported by the authors. MG = mitragynine; 7-OH = 7-hydroxymitragynine; SD = single dose; MD = multiple (daily) doses.

Study	Population and dose	Mitragynine	7-OH
Trakulsrichai 2015 [5]	9 chronic male users; tea, 6.25–23 mg MG; no washout	$t_{\max}$ 0.83 h; $t_{1/2}$ 23.2±16.1 h; bi-exponential; $C_{\max}$, AUC proportional to dose	not measured
Tanna 2022 [3]	5–6 healthy adults; 2 g leaf powder as tea	$t_{\max}$ 1 h; $t_{1/2}$ 45.3 h (median); two-compartment model fits best	$t_{\max}$ 1 h; $t_{1/2}$ 5.7 h; AUC ratio to MG 0.24
Mongar 2024 [4]	15–16 healthy adults; tea, 23.6 mg MG; ± itraconazole	$C_{\max}$ 159 ng/mL at 0.84 h; $t_{1/2}$ 8.7 h (72 h sampling); biphasic	$C_{\max}$ 12.8 ng/mL at 1.77 h; $t_{1/2}$ 7.4 h; CYP3A4-dependent
Huestis 2024 [1]	8–13 per arm, healthy adults; capsules 0.5–4 g leaf (6.65–53.2 mg MG); SD and 15 daily doses	$t_{\max}$ 1.0–1.3 h; $C_{\max}$ dose-proportional; median $t_{1/2}$ 3.7→42.9 h (SD), 14.6→61.2 h (MD); steady state 8–9 d	$t_{\max}$ 1.2–1.8 h; $t_{1/2}$ 1.7–4.7 h (SD); ratio to MG 0.2–0.3 (SD), 0.15–0.21 (MD)
Huestis 2026 [2]	Extract (39.5 % MG), 9.9–59.2 mg MG; SD and 15 MD	$C_{\max}$ 32.8–196 ng/mL; $t_{\max}$ 1.3–2 h; AUC < 1.4× proportional; higher exposure than leaf	$C_{\max}$ 6.8–30.5 ng/mL; $t_{\max}$ 1.3–2.3 h
Notario 2026 [6]	Model fitted (Bayesian, HMC) to Mongar 2024 data	linked parent–metabolite, 2+2 compartments, first-order absorption	formation fraction $F_m = 0.65$

2.1 Absorption

All studies agree that absorption of oral MG is fast: median $t_{\max}$ is 0.8–1.3 h after tea or capsules and 1.3–2.0 h at steady state or after extracts. 7-OH appears in plasma within 15 min [4] and peaks 0.2–0.9 h after MG,

as expected for a formed metabolite. Some participants showed secondary 7-OH peaks after a meal 4 h post-dose [1]. Tanna *et al.* noted that sparse early sampling limited the characterisation of absorption [3].

2.2 Distribution and elimination

Concentration–time profiles are bi-exponential: "an initial rapid descent after $C_{\max}$ for about 6 h and then a much slower terminal phase" [1]; see also [5, 3, 4]. Apparent volumes are very large (MG V_z/F 1 000–13 000 L), consistent with extensive tissue distribution of a lipophilic base. The terminal half-life reported for MG varies widely (8.7–68 h). Most of this spread is methodological:

- *Sampling duration and assay sensitivity.* The slow phase is visible only as long as concentrations stay above the limit of quantification. Studies that sample for 72 h [4] or give low doses report short half-lives, and studies that sample for 5–10 days at higher doses report long ones [1, 3]. Huestis *et al.* state that half-lives "were best reflected at higher doses due to the ability to measure low analyte concentrations longer" [2].
- *Possible saturation.* The same authors also raise possible enzyme saturation at higher doses [2]. AUC rose slightly more than dose-proportionally (1.18–1.42-fold over the dose range) [1].

The terminal phase carries little of the total exposure: in [1] $\mathrm{AUC}_{0\text{–}24}$ was already 61–80 % of $\mathrm{AUC}_{0\text{–}\infty}$ after one dose. Accumulation on once-daily dosing is therefore modest (MG $C_{\max}$ ratio 1.1–1.3, AUC ratio 1.6–1.9; 7-OH 0.9–1.0 and 1.0–1.3) [1].

2.3 Linearity and variability

$C_{\max}$ of both analytes increased dose-proportionally over 6.65–53.2 mg MG, and AUC increased slightly more than proportionally [1]; extracts gave less than 1.4× the proportional increase [2]. Trakulsrichai found linear kinetics over 6–23 mg [5]. A linear model is therefore a reasonable first approximation. Between-subject variability is large: at 0.5 g leaf, MG $C_{\max}$ ranged 1.8–61.5 ng/mL; at 4 g, 34–204 ng/mL [1]. Sources of variability include CYP3A4 activity [4, 2], food, and product composition.

2.4 Formulation

Exposure per mg of MG differs between preparations and studies: the tea of Mongar *et al.* gave $\mathrm{AUC}_{0\text{–}\infty}$ of 940 ng h/mL for 23.6 mg MG [4], whereas 26.6 mg in capsules gave 364 ng h/mL [1], and extracts gave more than leaf powder [2]. Because our users take plain leaf powder, the model is calibrated to the capsule study [1], which used the same material.

3 Model

3.1 Structure

We adopt the linked parent–metabolite model of Notario *et al.* [6], which describes both analytes with two compartments each and first-order absorption (Figure 1). Two empirical scale factors adapt it from tea to leaf powder (§4).

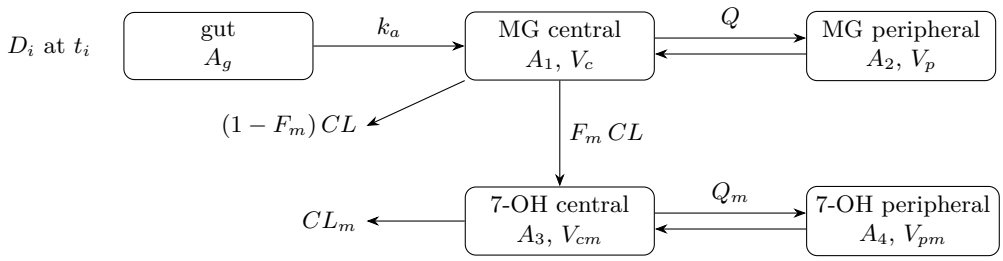


Figure 1: Model structure. Each dose D_i (mg MG) enters the gut at time t_i. A fraction F_m of MG clearance forms 7-OH. All parameters are apparent (divided by the unknown oral bioavailability).

With amounts A in μg and volumes in L (so that A/V is in ng/mL), the system is

$$\dot{A}_g = -k_a A_g, \tag{1}$$

$$\dot{A}_1 = k_a A_g - \frac{CL+Q}{V_c} A_1 + \frac{Q}{V_p} A_2, \qquad \dot{A}_2 = \frac{Q}{V_c} A_1 - \frac{Q}{V_p} A_2, \tag{2}$$

$$\dot{A}_3 = F_m \frac{CL}{V_c} A_1 - \frac{CL_m + Q_m}{V_{cm}} A_3 + \frac{Q_m}{V_{pm}} A_4, \qquad \dot{A}_4 = \frac{Q_m}{V_{cm}} A_3 - \frac{Q_m}{V_{pm}} A_4, \tag{3}$$

with plasma concentrations $C_{\mathrm{MG}} = A_1/V_c$ and $C_{\mathrm{OH}} = s_{\mathrm{OH}}\, A_3/V_{cm}$. A dose of g grams of powder contains $D = 1000\, f_{\mathrm{MG}}\, g$ mg MG ($f_{\mathrm{MG}} = 0.016$, i.e. 16 mg per gram) and adds $1000\, F_{\mathrm{rel}}\, D$ μg to A_g.

Table 2: Parameters: posterior medians of Notario *et al.* [6] (fitted to [4]), the scale factors fitted here, and derived quantities.

	Mitragynine		7-hydroxymitragynine		
k_a	1.99	h^{-1}	F_m	0.65	–
CL	17.85	L/h	CL_m	106.29	L/h
V_c	115.71	L	V_{cm}	50.61	L
Q	10.76	L/h	Q_m	8.52	L/h
V_p	355.64	L	V_{pm}	71.76	L
Fitted here (leaf powder)					
F_{rel}	0.277	–	s_{OH}	1.85	–
Derived					
absorption $t_{1/2}$	0.35	h			
α ($t_{1/2,\alpha}$)	0.2595	h^{-1} (2.67 h)	α_m ($t_{1/2}$)	2.2778	h^{-1} (0.30 h)
β ($t_{1/2,\beta}$)	0.0180	h^{-1} (38.54 h)	β_m ($t_{1/2}$)	0.1095	h^{-1} (6.33 h)

The MG disposition rate constants follow from $k_{10} = CL/V_c$, $k_{12} = Q/V_c$, $k_{21} = Q/V_p$ as $\alpha, \beta = \frac{1}{2}\left[(k_{10}+k_{12}+k_{21}) \pm \sqrt{(k_{10}+k_{12}+k_{21})^2 - 4k_{10}k_{21}}\,\right]$, and analogously for 7-OH. The resulting MG distribution half-life of 2.67 h and terminal half-life of 38.54 h agree with the "fast fall for about 6 h, then slow tail" pattern and with the long half-lives reported by studies that sampled for 5–10 days (Table 1). 7-OH's own disposition is fast (0.30 h and 6.33 h), so its plasma level is *formation-rate-limited*: after the first few hours it follows the MG curve with a roughly constant ratio. The short 7-OH half-lives reported clinically (1.7–5.7 h) reflect the fast distribution phase of the parent, as the metabolite falls below the limit of quantification before the slow tail is reached.

3.2 Direct 7-OH in the powder

The users' product contains 0.001 % 7-OH (0.01 mg per gram) against 16 mg MG per gram. Even assuming complete bioavailability of ingested 7-OH, its contribution to 7-OH exposure (same clearance CL_m) is at most $0.01/(s_{\mathrm{OH}}\, F_{\mathrm{rel}}\, F_m \cdot 16) = 0.19\,\%$ of the 7-OH formed from MG. It is therefore neglected.

3.3 Analytical solution and superposition

The system $\dot{\mathbf{A}} = \mathbf{K}\mathbf{A}$ is linear and time-invariant, and $\mathbf{K}$ is lower block-triangular with five distinct real eigenvalues $-\lambda_k \in \{-k_a, -\alpha, -\beta, -\alpha_m, -\beta_m\}$. The response to a unit dose is therefore a sum of exponentials,

$$h_{\mathrm{MG}}(\tau) = \sum_{k=1}^{5} a_k e^{-\lambda_k \tau}, \qquad h_{\mathrm{OH}}(\tau) = \sum_{k=1}^{5} b_k e^{-\lambda_k \tau}, \qquad \tau \geq 0, \tag{4}$$

obtained from the eigen-decomposition $\mathbf{K} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$: $\mathbf{A}(\tau) = \mathbf{V} e^{\mathbf{\Lambda}\tau} \mathbf{V}^{-1} \mathbf{A}(0)$. Here h is the response to 1 mg of MG entering the gut. For a dosing history $\{(t_i, g_i)\}$ (grams of powder at times t_i, i.e. $D_i = 16\, g_i$ mg MG) the concentrations are exact superpositions,

$$C_{\mathrm{MG}}(t) = F_{\mathrm{rel}} \sum_{i:\, t_i \leq t} D_i\, h_{\mathrm{MG}}(t - t_i), \qquad C_{\mathrm{OH}}(t) = s_{\mathrm{OH}} F_{\mathrm{rel}} \sum_{i:\, t_i \leq t} D_i\, h_{\mathrm{OH}}(t - t_i). \tag{5}$$

No numerical integration is needed in the app; the coefficients (per gram of the users' powder, already including F_{rel}, f_{MG} and s_{OH} for c_k below) are listed in Table 7. The coefficients of each analyte sum to zero, so $h(0) = 0$.

4 Calibration and validation

4.1 Procedure

The Notario parameters reproduce the tea study they were fitted to (model vs. observed [4]: MG C_{max} 151.9 vs. 159 ng/mL, t_{max} 1.20 vs. 0.84 h; 7-OH C_{max} 14.5 vs. 12.8 ng/mL, t_{max} 1.85 vs. 1.77 h). For leaf-powder capsules, exposure per mg MG is lower and the 7-OH:MG ratio is higher (Table 1). We therefore estimated a relative apparent bioavailability F_{rel} (scaling both analytes) and a metabolite scale s_{OH} (scaling 7-OH only) by least squares on the log scale, using the 16 single-dose mean values of Huestis *et al.* [1] (C_{max} and $AUC_{0-\infty}$ of both analytes at four doses). The time-course parameters were not refitted. F_{rel} absorbs everything that makes the capsule cohort's exposure per mg lower than the tea cohort's (formulation, population, assay), and s_{OH} absorbs the higher metabolite ratio. Neither has a mechanistic interpretation.

Result: $F_{rel} = 0.277$, $s_{OH} = 1.85$, root-mean-square log error 0.382 (a typical deviation of $\times e^{0.382}$). The residual pattern is systematic (Table 3). (i) Peaks are 17–30 % lower than observed at every dose, i.e. the tea-derived time course is slightly flatter than the capsule data. (ii) AUC is over-predicted at low doses (about 2× at 6.65 mg) and slightly under-predicted at 53.2 mg, because observed AUC grows more than proportionally with dose while a linear model cannot. Both deviations are well within the between-subject range of the data.

Table 3: Calibration: single dose, observed (means, Huestis 2024 [1]) vs. model. C_{max} in ng/mL; $AUC_{0-\infty}$ in ng h/mL.

	MG C_{max}		MG AUC		7-OH C_{max}		7-OH AUC	
mg MG	obs	model	obs	model	obs	model	obs	model
6.65	17.1	11.9	52.8	103	3.6	2.1	11.2	21
13.3	32.1	23.7	136	207	6.6	4.2	28.8	42
26.6	57	47.5	364	413	10.8	8.4	58.2	83
53.2	125	95.0	908	827	22.7	16.8	136	167

4.2 Validation against repeated dosing

The calibrated model was then used to predict day 15 of once-daily dosing, data not used in the fit (Table 4, Figure 2b). Peaks and AUCs are predicted within a factor of about 2 (the largest deviation is the 7-OH AUC at the lowest dose), and MG troughs within 20 % at the three lower doses. At 53.2 mg the trough is under-predicted (12.4 vs. 20.5 ng/mL), consistent with the longer half-life observed at high doses. The model reproduces the modest observed accumulation: the steady-state peak of the combined index is 1.145 times the single-dose peak (observed MG C_{max} ratio 1.1–1.3).

Table 4: Validation: day 15 of daily dosing, observed (means [1]) vs. model. Trough = concentration 24 h after the 15th dose.

	MG $C_{max,ss}$		MG trough		MG AUC_{τ}		7-OH $C_{max,ss}$		7-OH AUC_{τ}	
mg MG	obs	model	obs	model	obs	model	obs	model	obs	model
6.65	21.4	13.4	1.4	1.55	85.1	103	3.2	2.4	10.1	21
13.3	33.5	26.8	2.74	3.09	175	207	5.9	4.8	28	42
26.6	74.6	53.5	7.7	6.19	457	413	11.3	9.6	67.8	83
53.2	143	107.0	20.5	12.37	958	826	21.8	19.3	133	167

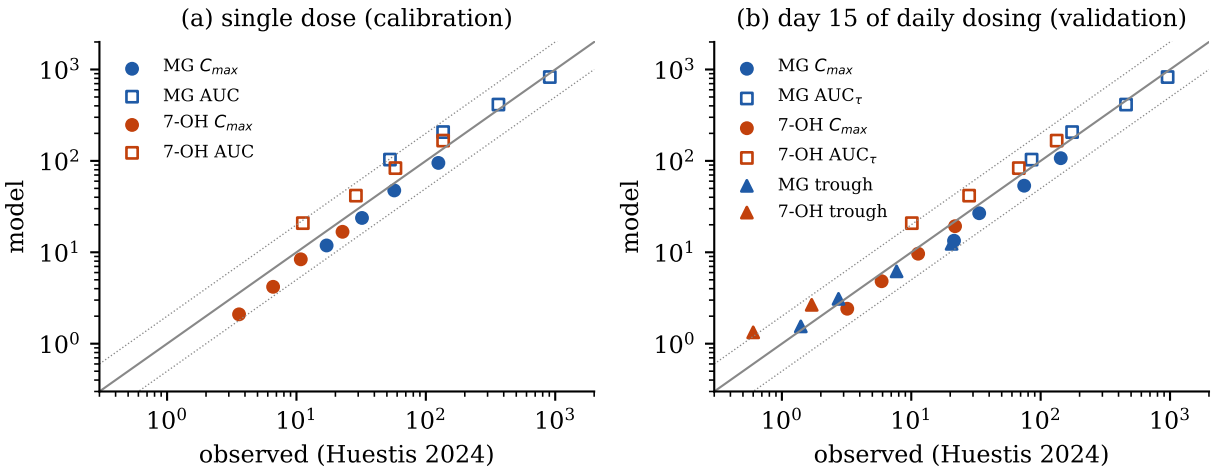


Figure 2: Observed vs. model. (a) Single doses (calibration data). (b) Day 15 of daily dosing (validation, not fitted). Solid line: identity; dotted lines: factor 2.

5 A combined activity index

The app shows one combined line rather than two concentrations. We define a potency-weighted MOR-equivalent concentration

$$E(t) = \frac{C_{\mathrm{MG}}(t)}{M_{\mathrm{MG}}} + w\,\frac{C_{\mathrm{OH}}(t)}{M_{\mathrm{OH}}} \quad [\mathrm{nM}], \qquad w = \frac{EC_{50,\mathrm{MG}}}{EC_{50,\mathrm{OH}}} = \frac{339}{34.5} = 9.83, \tag{6}$$

with molar masses $M_{\mathrm{MG}} = 398.5$ and $M_{\mathrm{OH}} = 414.5\,\mathrm{g/mol}$ (concentrations in ng/mL $\times 1000/M$ give nM). E is the MG concentration that would give the same receptor stimulation in the low-occupancy (linear) range. Underlying assumptions: (i) equal unbound fractions (>90 % bound for both [11]); (ii) equal access to the receptor compartment (brain distribution kinetics are not modelled); (iii) potency ratio from one *in vitro* assay [8]; (iv) the difference in $E_{\max}$ (34 vs. 47 %) and receptor saturation are ignored; (v) non-opioid targets are ignored.

With $w = 9.83$, 7-OH accounts for 66 % of the area under E after a single dose (63 % at the peak), in line with the finding that formed 7-OH mediates much of MG's opioid effect [9]. Because 7-OH follows the MG curve closely (formation-limited), the *shape* of E is insensitive to w (Figure 6a): it differs from MG alone mainly by a slightly later peak.

6 Relative scale shown to the user

Absolute concentrations are not shown. The displayed level is

$$L(t) = 100\,\frac{E(t)}{R \cdot G_0} \quad [\%], \tag{7}$$

where G_0 is the user's plan baseline (g/day, field `baselineG`) and $R = 75.55\,\mathrm{nM}$ per g/day is the steady-state *daily mean* of E when 1 g/day is taken. Because the model is linear, the steady-state mean depends only on the daily amount, not on how it is split ($\bar{E} = \frac{1}{24}\,G_0 \int_0^\infty h_E\,d\tau = \frac{G_0}{24}\sum_k c_k/\lambda_k$). Hence **100 % is the average level at the start of the taper**, and a plan that halves the daily amount brings the daily mean to 50 %. Peaks rise above 100 %: a single baseline-sized dose from a drug-free state peaks at 248 %, and taking the whole baseline once daily at steady state gives peaks of 284 % (accumulation factor 1.145). Splitting the amount into several doses gives lower peaks (Figure 4b).

Table 5: Model time course after a single dose, as % of each curve's own peak.

hours after dose	1	2	4	6	8	12	24	48	72	168
MG	99	91	58	37	24	12	5	3	2	0
7-OH	80	100	71	46	31	16	6	3	2	0
combined E	88	98	67	43	29	15	6	3	2	0

After stopping from steady state (baseline taken once daily), L falls to 37.5 % one day after the last dose, 23.3 % after 2 days, 15.1 % after 3 days, 2.7 % after 7 days and 0.1 % after 14 days (measured from the last dose, as % of the baseline daily mean). A dose 7 days old contributes 0.38 % of its peak and a dose 14 days old 0.02 %, so the app only needs the last 14 days of doses.

7 Amount remaining in the body

Alongside the relative level, the app shows how many milligrams of each analyte are still in the body. The model parameters are *apparent* (divided by the unknown absolute oral bioavailability F; no intravenous human data exist), so true amounts cannot be computed. We therefore report the *dose-equivalent amount not yet eliminated*, i.e. the amounts the model holds when the whole ingested dose is treated as absorbed (F_{rel} not applied):

$$M_{\mathrm{MG}}(t) = \sum_i D_i \left[A_1 + A_2\right]_{\mathrm{unit}}(t - t_i), \qquad M_{\mathrm{OH}}(t) = s_{\mathrm{OH}} \sum_i D_i \left[A_3 + A_4\right]_{\mathrm{unit}}(t - t_i), \tag{8}$$

where $[\cdot]_{\mathrm{unit}}$ is the response to 1 mg in the gut. Unabsorbed drug in the gut is not counted. This is an *upper estimate* of the amount actually in the body, and it is the natural reading of "how much of what I took is still in me". Because most MG resides in the peripheral compartment ($V_p \approx 3V_c$), M_{MG} changes much more smoothly than plasma concentration. Example (70 kg adult, the population of the calibration studies; 6 g/day in three doses, day 14): M_{MG} ranges 87–121 mg and M_{OH} 3.8–6.8 mg over the day, against 96 mg MG ingested per day. The model is not scaled for body weight. Coefficients are listed in Table 6.

Table 6: Coefficients for the amounts still in the body, $M(\tau) = \sum_k d_k e^{-\lambda_k \tau}$, in mg per gram of the users' powder.

k	λ_k (h^{-1})	d_k MG (mg/g)	d_k 7-OH (mg/g)
1	0.017984	+7.03624	+0.19781
2	0.109472	0	+1.48270
3	0.259527	+10.38122	-0.29434
4	1.990000	-17.41746	-10.66058
5	2.277782	0	+9.27441

8 Model behaviour

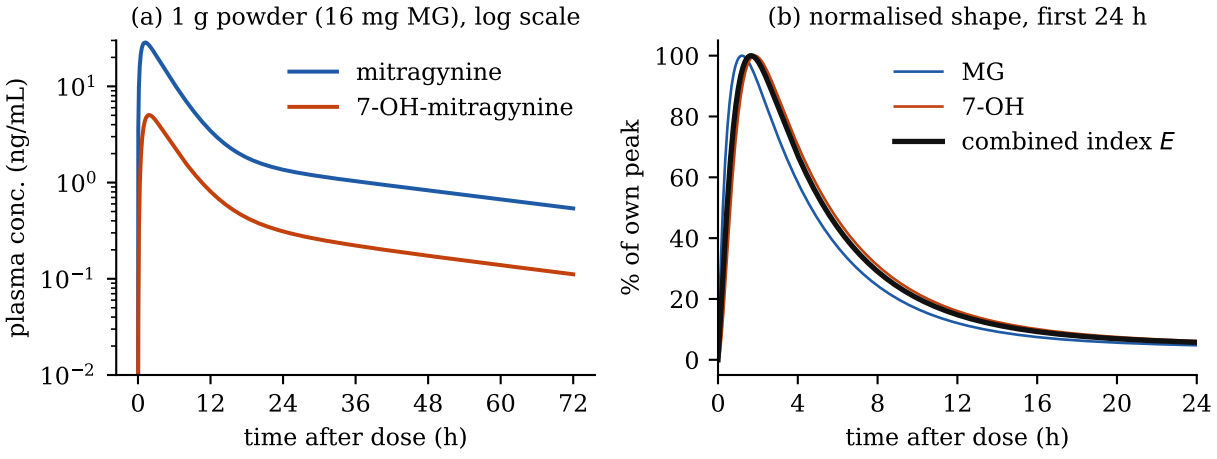


Figure 3: Single dose of 1 g powder (16 mg MG). (a) Absolute model concentrations, log scale: fast distribution phase, then a terminal phase with $t_{1/2} \approx 38.54$ h; 7-OH runs parallel to MG. (b) Normalised shapes over 24 h.

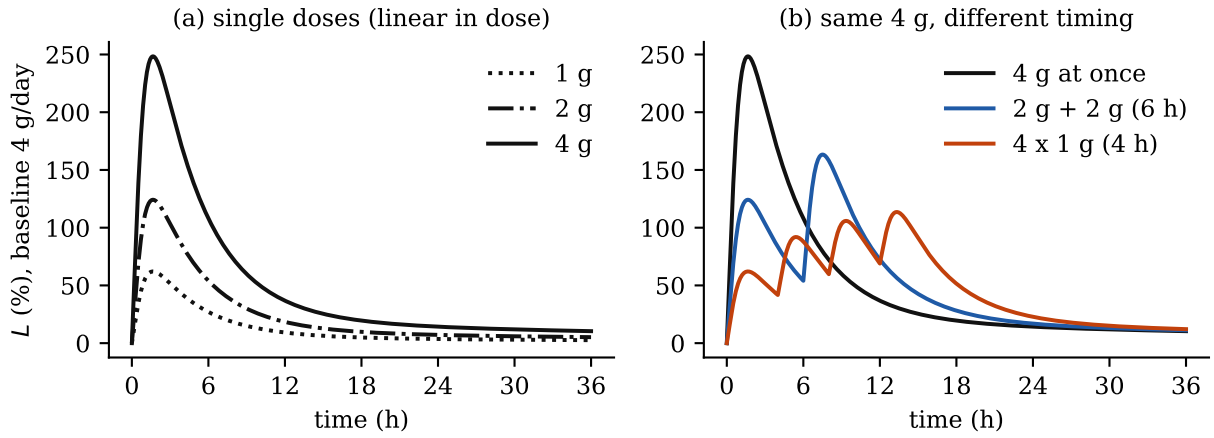


Figure 4: Dose and timing responsiveness, shown as L for a user with baseline $G_0 = 4\,\mathrm{g/day}$. (a) Peaks scale linearly with dose. (b) The same daily amount split into more doses gives a lower, broader curve.

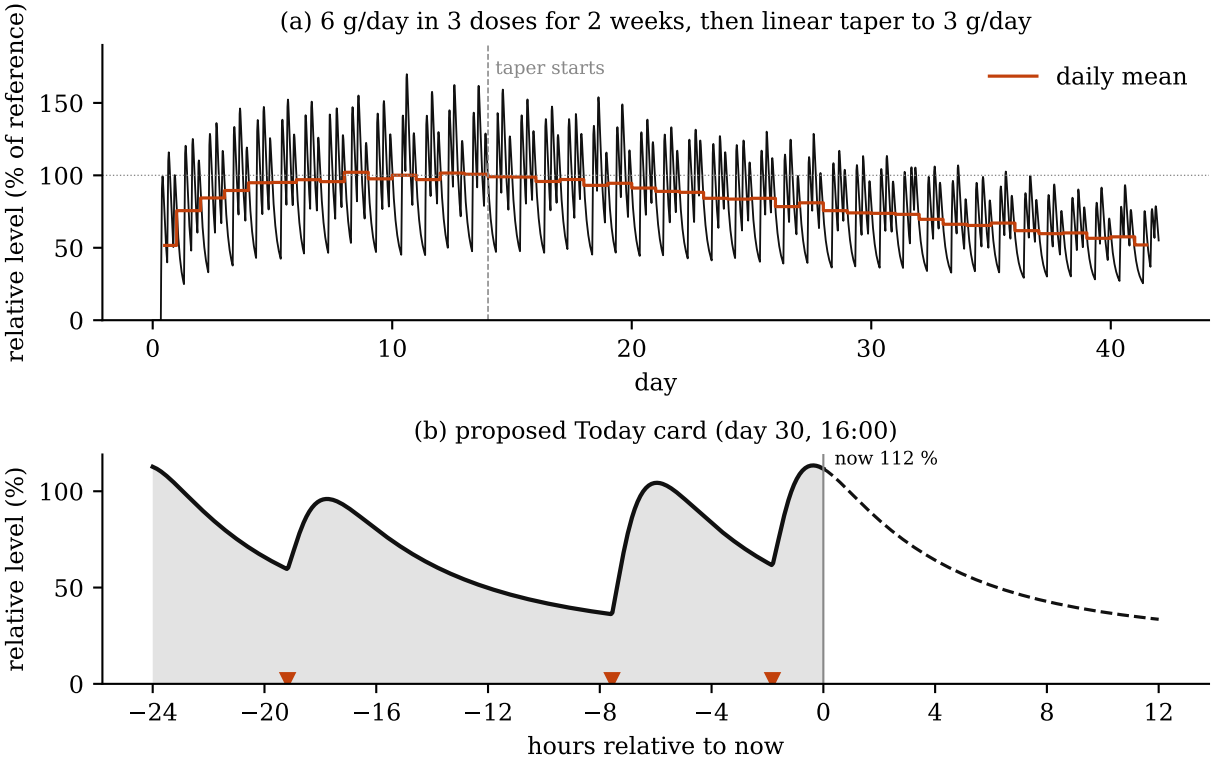


Figure 5: Simulated user (illustrative, random dose times): baseline 6 g/day in three doses for two weeks, then a linear taper to 3 g/day. (a) $L(t)$ over six weeks and its daily mean, which starts at 100 % and ends near 50 %. (b) The proposed "Today" card: last 24 h (solid), the moment "now", and the projection if no further dose is taken (dashed); triangles mark doses.

9 Uncertainty and limitations

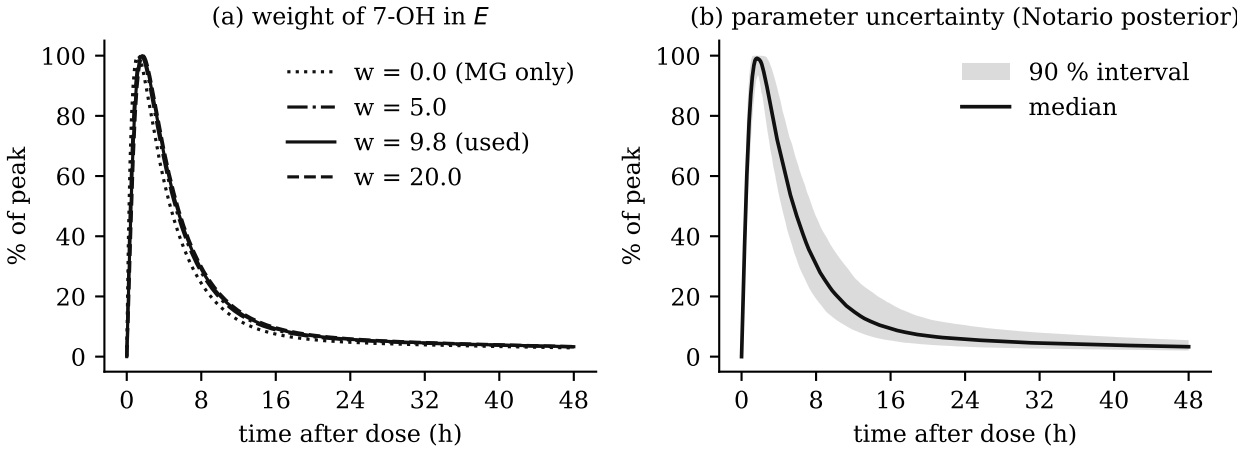


Figure 6: (a) Shape of E for different 7-OH weights w. (b) Uncertainty of the population-average shape from the parameter posterior (log-normal approximation of [6], 400 draws). At 12 h the level is 15 % of peak (90 % interval 9–27 %), at 24 h 6 % (3–10 %).

1. **Population average, not the individual.** Between-subject variability in exposure is 5–10-fold [1] and is not represented. Figure 6b shows only the uncertainty of the *average* curve. Normalising to the user's own baseline removes the overall scale, but not individual differences in shape (e.g. CYP3A4 activity).
2. **Linearity.** AUC rises up to ∼1.4× faster than dose [1, 2], and the high-dose trough is under-predicted (Table 4). The relative display slightly understates how much a large dose adds compared with a small one.
3. **Calibration source.** One capsule study, with fasting. Our users' preparation (toss-and-wash, with or without food) may absorb differently. Food delays absorption and can cause secondary peaks [1].
4. **Chronic use.** Most participants were healthy adults, not daily long-term users. In chronic users, Trakulsrichai reported $t_{1/2} \approx 23$ h [5], within the range modelled.
5. **Product composition.** 1.6 % MG is taken as fixed; real batches vary. Other alkaloids are ignored.
6. **Interactions.** CYP3A4 inhibitors (e.g. itraconazole, clarithromycin, grapefruit) lower 7-OH and raise MG [4]; inducers (e.g. rifampicin, St John's wort) are expected to do the opposite. Not modelled.
7. **Concentration is not effect.** Tolerance, receptor adaptation and withdrawal are not modelled; withdrawal symptoms do not map onto a plasma level. A low displayed level does not mean that a dose is needed.
8. **Not applicable** to extracts (higher exposure per mg [2]) or to semi-synthetic 7-OH products.

10 Implementation specification

Table 7: Closed-form coefficients for $h(\tau) = \sum_k \mathrm{coef}_k\, e^{-\lambda_k \tau}$ (τ in h). a_k: MG in ng/mL per mg MG reaching the gut (multiply by $16\,F_{\mathrm{rel}}$ per gram); b_k: 7-OH (unscaled) likewise; c_k: combined index E in nM per gram of the users' powder (all factors included).

k	λ_k (h^{-1})	$t_{1/2,k}$ (h)	a_k	b_k	c_k
1	0.017984	38.54	+0.443066	+0.049497	+14.56008
2	0.109472	6.33	0	+0.051653	+10.04373
3	0.259527	2.67	+9.433512	+1.005442	+300.59118
4	1.990000	0.35	-9.876578	-7.829166	-1632.38407
5	2.277782	0.30	0	+6.722574	+1307.18908

1. **Module** `src/lib/pk.ts`: constants λ_k, c_k, R and the amount coefficients of Table 6 (copied from `ts_constants.json`); a pure function `level(doses, t, baselineG)` implementing $L(t) = 100 \sum_i g_i \sum_k c_k e^{-\lambda_k (t - t_i)} /$

(with $R = 75.55$) over doses with $0 \le t - t_i \le 14$ d, and a helper that returns a series on a 10-min grid for charting.

2. **Data.** The user's own doses for the last 14 days (`takenAt`, `grams`) and the plan's `baselineG`. Runs entirely in the browser; no database change. Other users' curves are never computed (their dose times are private).
3. **Today card** (below the list of today's doses). Milligrams of MG and 7-OH still in the body (§7), current L (rounded to 5 %; it can exceed 100 % shortly after a dose), a 24 h history, dose markers and a dashed 12 h projection with no further doses. Without an active plan there is no baseline, so the card shows the amounts and the curve without a percentage scale. Doses logged at a later time today appear when their time comes.
4. **Tests** (`pk.test.ts`): (i) the sums of exponentials agree with a numerical ODE solution; (ii) the single-dose and day-15 values in Tables 3 and 4; (iii) superposition and linearity; (iv) the daily mean of L is 100 % at steady state on the baseline amount, however it is split.
5. **Wording.** Always labelled as a rough population-average estimate, not a measurement, and with no advice derived from it.

Reproducibility

All numbers, tables and figures in this report are generated by `docs/pk-model/model.py` (NumPy, SciPy, Matplotlib), which writes `numbers.tex`, `numbers.json` and `fig/*.pdf`.

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